

SuperApp

A full pipeline Data Cleaning & Exploratory Data Analysis Application

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Problem

- Time consumed when converting invoice into usable excel dataset
- Highly repetitive manual data cleaning and naming/grouping convention
- No specific guideline for naming/grouping convention

Solution

- Seamless full pipeline Invoice Converter → Data Pre-processing → Data Visualization
- A 3-in-1 simple, simple & intuitive application usable by everyone regardless of their technical skill
- 2 reports output tailored for management's needs (Yearly Overview Report & Monthly Detailed Report)
- Reducing manual data entry which can save hours of work

Scope of Project

- Limited to WMG's Internal use only

Introducing: **SuperApp**

A seamless full pipeline 3-in-1 simple yet intuitive python-native Application made for the needs of West Malayan Group's Management to analyze sales data by yearly or monthly.

The application is highly tailored to fulfill the requirements and to make it easy for everyone including non-technical employee to use it.



Key features of SuperApp

Auto Invoice Converter (Step 1)

The invoices generated and extracted from SQL Accounting are still in the 'raw' format, hence it needs to be converted to usable & workable excel dataset on this stage.

Seamless Data Cleaning and Grouping (Step 2)

The 2nd part of the application handles the basic data cleaning (spacing normalization) and outlet grouping. It uses mapping.csv to group each outlet to their belonging group. User can also add, edit, and delete the new or existing outlet names. At this stage user can export the converted dataset for other purpose.

Auto Exploratory Data Analysis (Step 3)

The program can generate both overview and detailed visualization. User will be able to choose which report to export (Yearly Overview Report / Monthly Detailed Report) . User can also import external dataset to be visualized on this stage.

Limitations and Suggestions

- SuperApp is in its earliest version. In the event that the management requires update of the application, change the scope of the project, future developers need to edit/update the source code, and later package it again using PyInstaller.
- SuperApp is limited to current/existing dataset analysis, suppose the management wants to see data forecasting, future developers are free to add forecasting function in the source code.
- SuperApp Data Cleaning function is limited to space normalization and outlet grouping. This is due to the relatively 'clean' WMG's dataset. To make sure and double check, developer added a function called 'export dataset' in the event that user needs to perform data accuracy & integrity check. Also note that WMG's dataset contains lots of (outlet) name inconsistency, suppose high-consistency analysis is needed, please perform manual data cleaning.
- CS Brothers outlet naming and grouping is hard coded, because they're using separated column for their branches code. Please proceed to check/update [cleaning.py](#)
- Econsave outlet naming and grouping is also hard coded, because they're using special numerical code in front of the actual name for their branches code. Please proceed to check/update [cleaning.py](#)

Limitations and Suggestions

- SuperApp is wrapped in Tkinter GUI (Graphical User Interface) for simplicity and ease of use, suppose in the future you want to make it “fancier” feel free to overhaul the GUI.
- SuperApp cannot choose/differentiate LKA&IKA, it's solely depends on the input dataset. Suggestions; future dev creates another grouping to differentiate between LKA & IKA outlets.
- SuperApp is limited to WMG's Internal use only.

Source Code

<https://github.com/jayy1705/WMG-SuperApp>

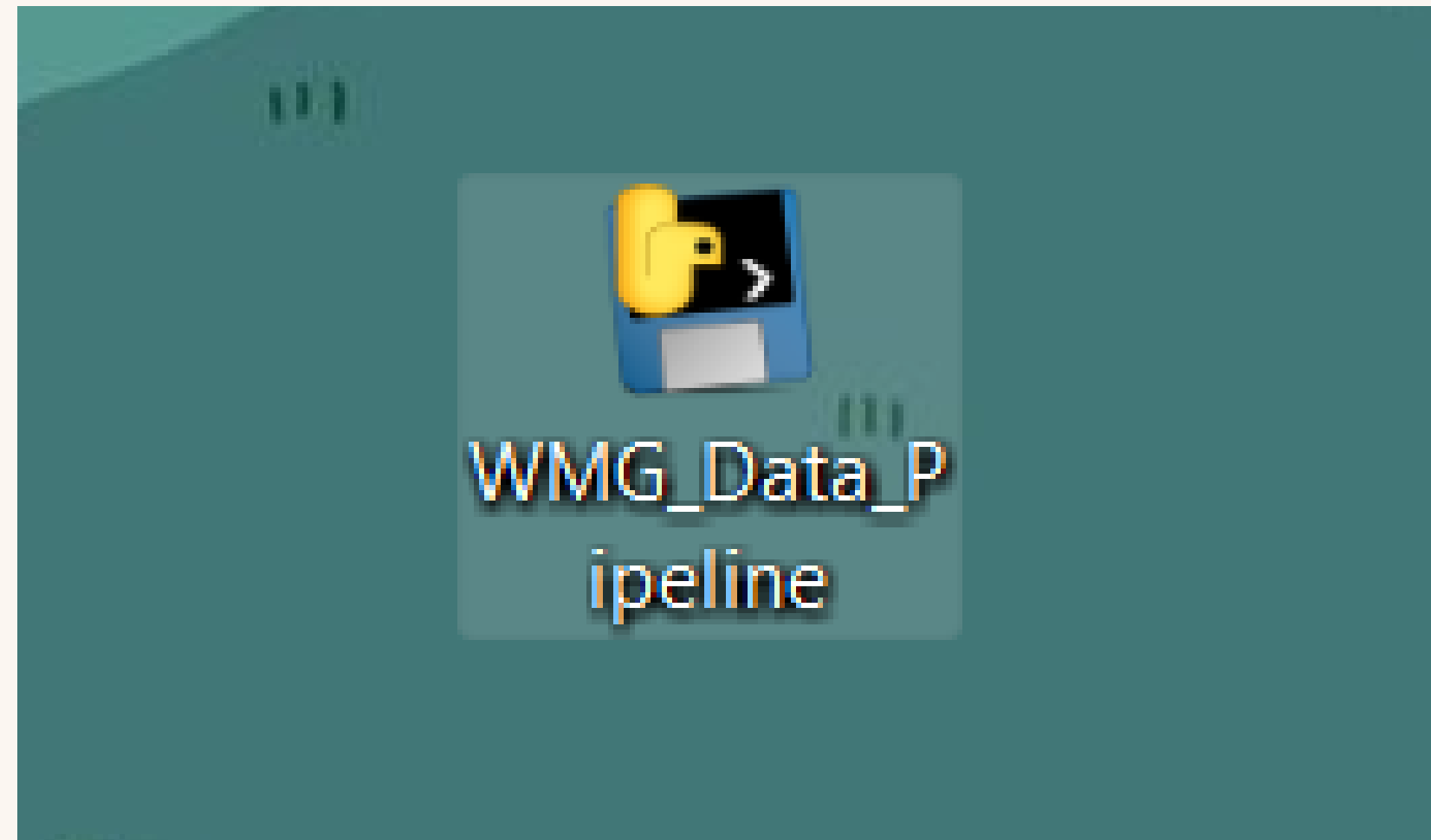
For the application only, please download dist/WMG_SuperApp.exe

Thank you & All the best !

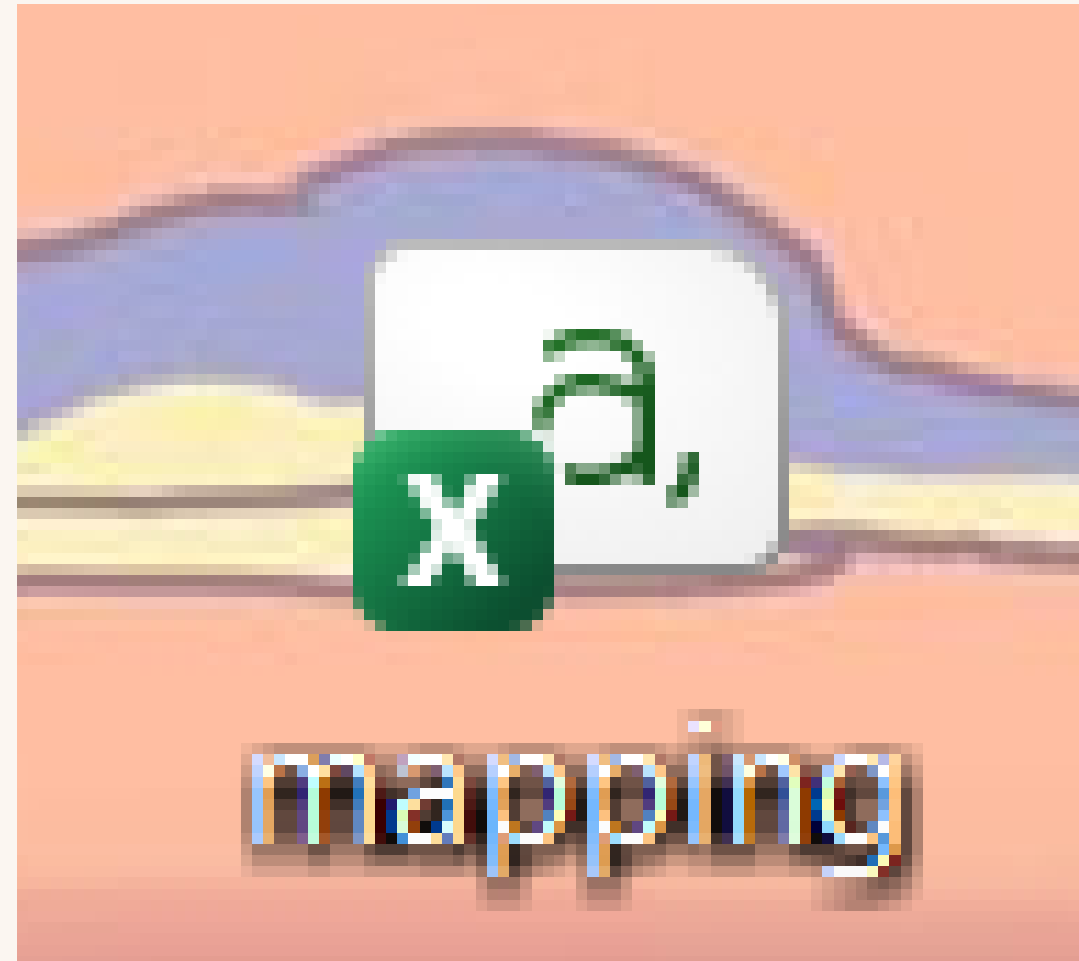


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Appendix

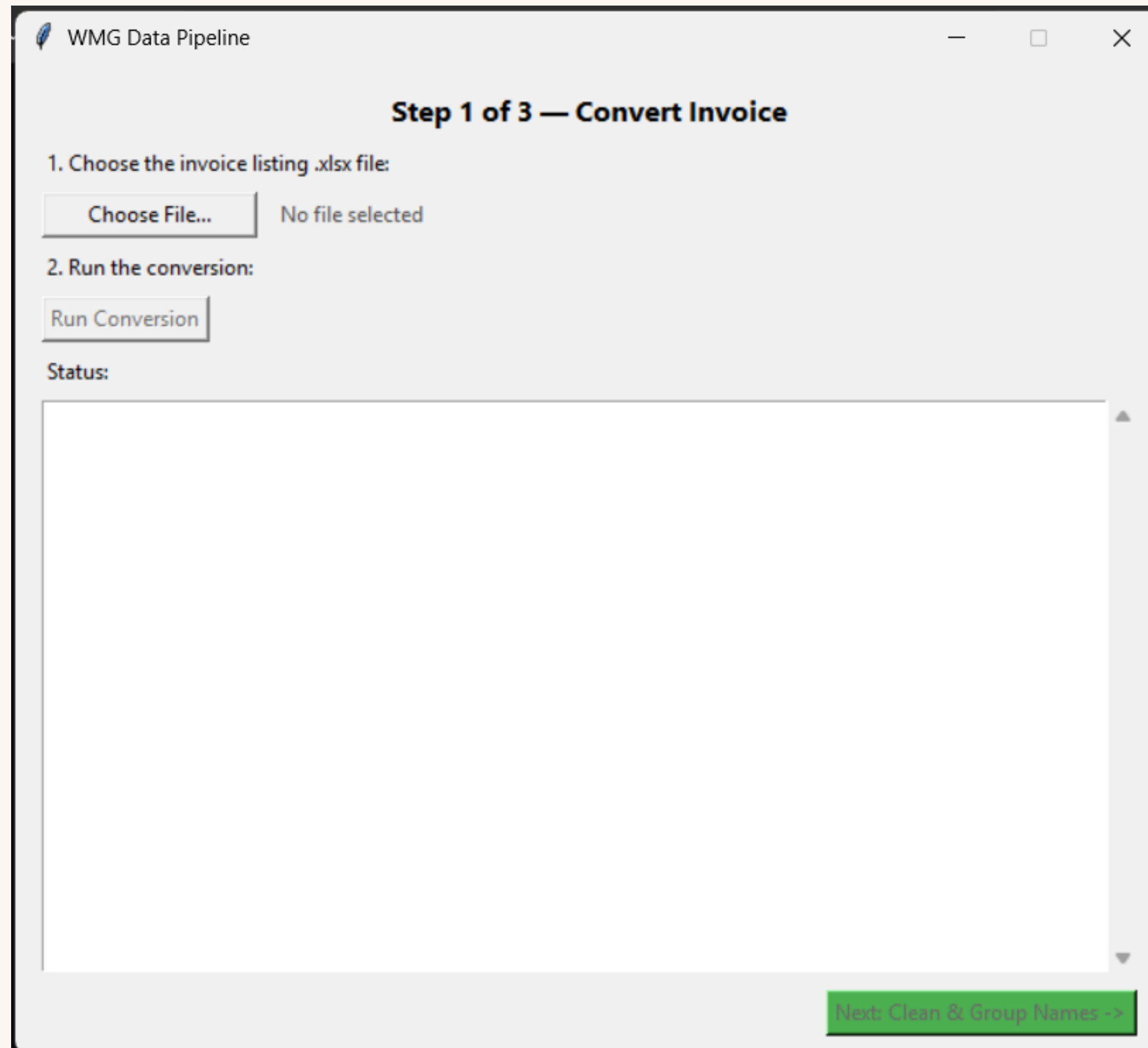


Double click **WMG_SuperApp** and run the application

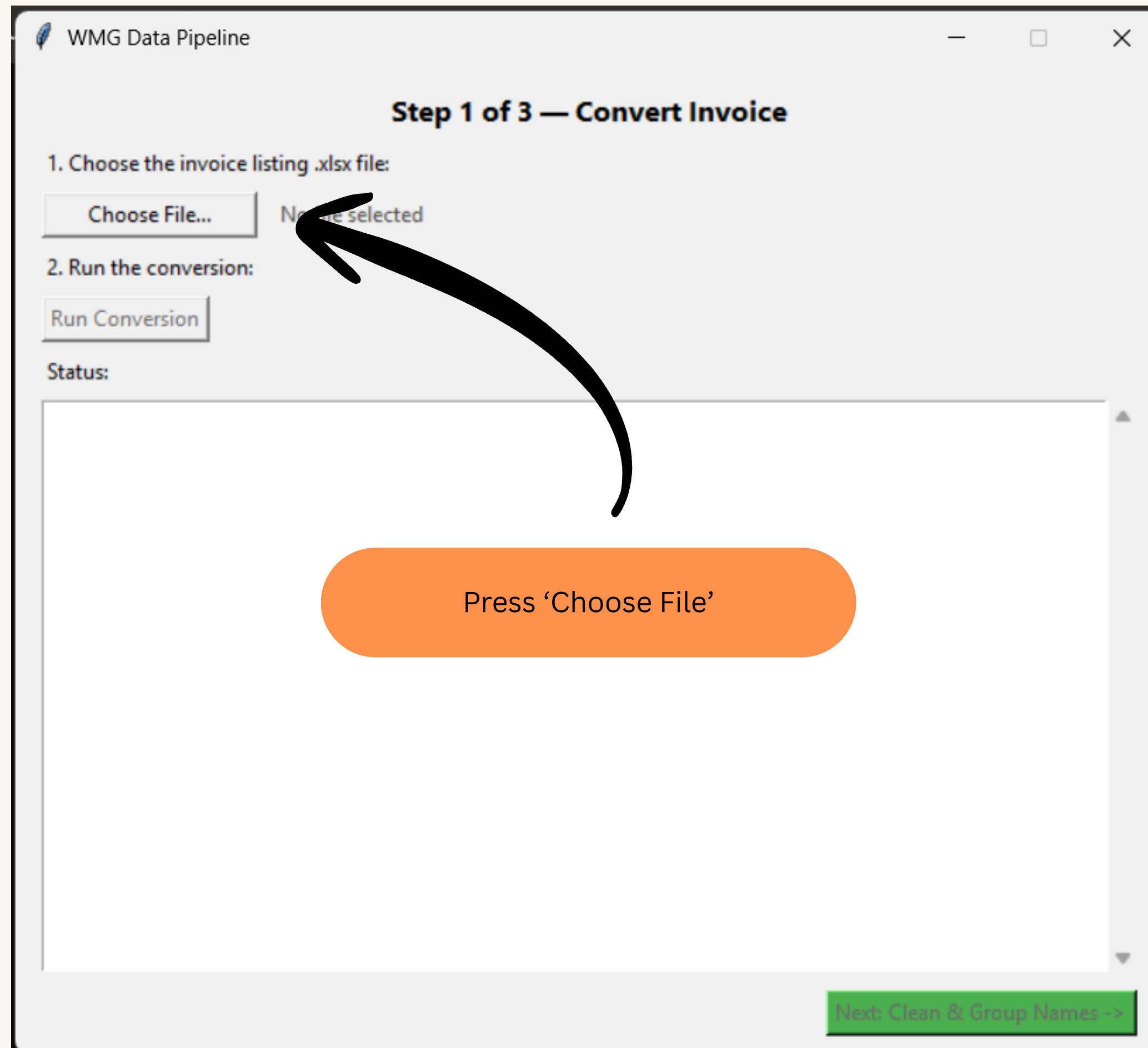


After the first run, you will notice that mapping.csv is created in the same folder.

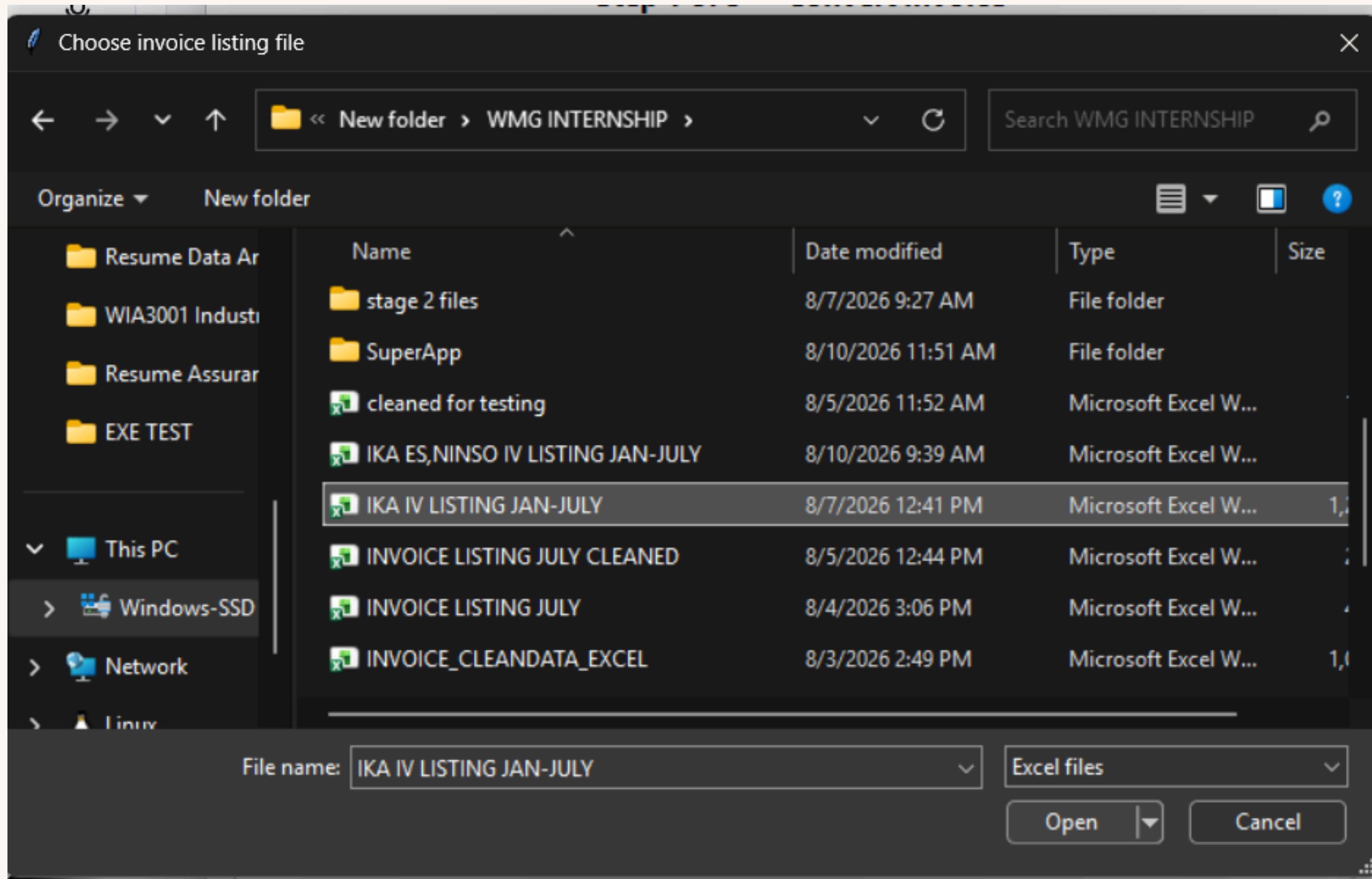
This mapping.csv is used to map the groupings, you can directly edit inside the .csv file



This is how it should look like when you open the app



This is how it should look like when you open the app



Select the 'raw' Invoice file

WMG Data Pipeline

Step 1 of 3 — Convert Invoice

1. Choose the invoice listing .xlsx file:

IKA IV LISTING JAN-JULY.xlsx

2. Run the conversion:

Status:

Selected: C:/New folder/WMG INTERNSHIP/IKA IV LISTING JAN-JULY.xlsx

Click 'Run Conversion'
it should take around 20-30 seconds

Step 1 of 3 — Convert Invoice

1. Choose the invoice listing .xlsx file:

Choose File...

IKA IV LISTING JAN-JULY.xlsx

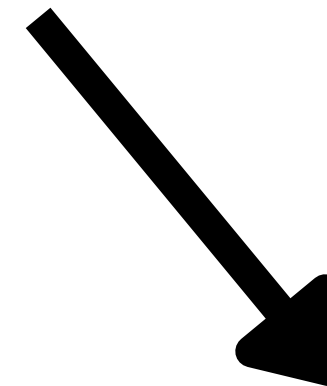
2. Run the conversion:

Run Conversion

Status:

```
Selected: C:/New folder/WMG INTERNSHIP/IKA IV LISTING JAN-JULY.xlsx  
Running conversion...  
Done. 11274 line items across 1683 invoices.  
All rows recognized, all invoice totals reconciled. Looks clean.
```

Click 'Next: Clean & Group Names'



Next: Clean & Group Names ->

Step 2 of 3 — Clean & Group Names

Received 10891 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Export Cleaned & Grouped Data...

Status

Cleaning complete. 10891 rows processed.
Missing values: 0 total
Duplicate rows (fully identical): 0
All names matched the mapping.

Click 'Run Cleaning & Grouping'

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Next: Analyze ->

Step 2 of 3 — Clean & Group Names

Received 10891 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Export Cleaned & Grouped Data...

Status

```
Cleaning complete. 10891 rows processed.  
Missing values: 0 total  
Duplicate rows (fully identical): 0  
All names matched the mapping.
```

If it says “All names matched the mapping”, you are clear to proceed to analyze

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Next: Analyze ->

Step 2 of 3 — Clean & Group Names

Received 10891 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Export Cleaned & Grouped Data...

Status

Cleaning complete. 10891 rows processed.
Missing values: 0 total
Duplicate rows (fully identical): 0
All names matched the mapping.

If not, click on 'Manage Customer Names..'

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Next: Analyze ->

Manage Customer Names

Raw name (as it appears in your data)	Group (canonical customer)
AEON (KL RDC)	AEON
BILLION MART (TANJUNG CHAT) SDN. BHD.	BILLION MART (TANJUNG CHAT) SDN. BHD.
BORONG DIN AS CASH & CARRY (ALOR JANGGUS)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY (PADANG BESAR)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY (SERIAB)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SB - PENDANG	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (JEJAWI)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KANGAR KAPITOL)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KANGAR)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (KUALA PERLIS)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (PAUH)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (POKOK SENA)	BORONG DIN AS CASH & CARRY
BORONG DIN AS CASH & CARRY SDN BHD (SIMPANG EMPAT)	BORONG DIN AS CASH & CARRY
PASARAYA BORONG DIN AS CASH & CARRY SDN BHD (KUALA NERANG)	BORONG DIN AS CASH & CARRY
PASARAYA BORONG DIN AS SD BHD (AYER HITAM)	BORONG DIN AS CASH & CARRY
CB FROZEN FOOD SDN BHD - DC	CB FROZEN FOOD SDN BHD - DC
CS BROTHERS SDN BHD	CS BROTHERS
CS GROCER SDN BHD (KAJANG REKO)	CS GROCER
CS GROCER SDN BHD (PUCHONG)	CS GROCER

Add New... Edit Selected... Delete Selected Cancel Save & Close

File: C:\Users\User\Desktop\mapping.csv

A new pop-up window will appear, this is where you can add/edit the grouping once u edited it, click 'Save & Close'

Step 2 of 3 — Clean & Group Names

Received 11274 rows from Step 1.

Run Cleaning & Grouping

Manage Customer Names...

Status

```
Cleaning complete. 11274 rows processed.  
All names matched the mapping.  
Customer mapping updated and reloaded.  
Note: re-run cleaning to apply the updated mapping.  
Cleaning complete. 11274 rows processed.  
1 name(s) not in the mapping (grouped as 'Other'):  
  - AEON BIG  
Customer mapping updated and reloaded.  
Note: re-run cleaning to apply the updated mapping.
```

Click on 'Run Cleaning & Grouping' one more time
now you are ready to analyze and generate the
report

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Next: Analyze ->

Step 3 of 3 — Analyze

Step 3: Analyze

Analyzing 10891 rows from Step 2.

Import External File to Analyze...

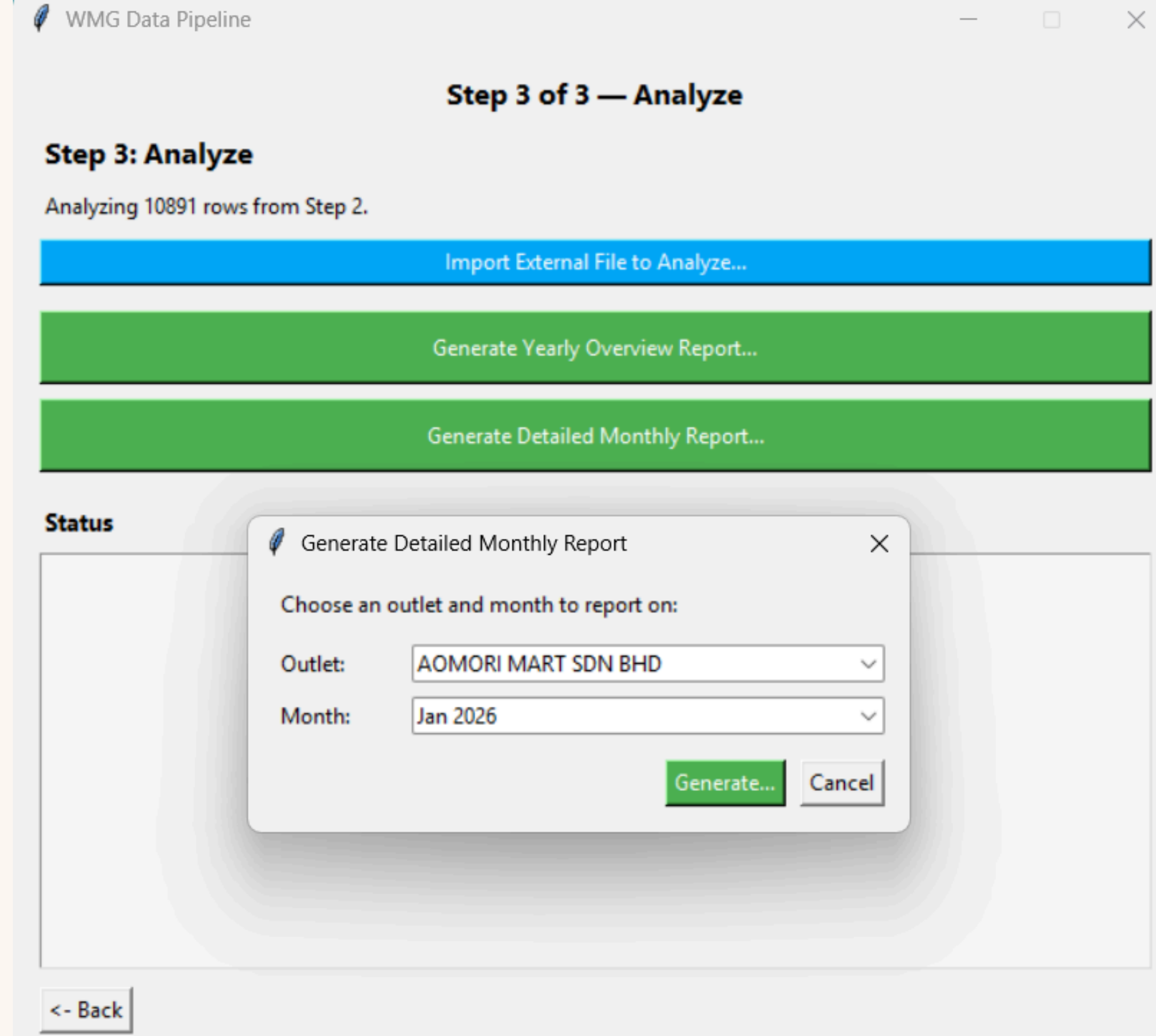
Generate Yearly Overview Report...

Generate Detailed Monthly Report...

Status

here, you will have 3 options, 'Generate Yearly Overview Report..' | 'Generate Detailed Monthly Report..', and 'Import External File to Analyze..' if you wish to use external file

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in ‘Generate Monthly Report..’ You will have the option to choose which month and which outlet

Finally, save the report and you can directly view it as PDF file